

Zomato Dataset Analysis

Internspark Internship Project

Project details	Notebook output
Notebook output	56,252 rows × 13 columns
After duplicate removal	40,549 rows
Unique restaurant names	11,914
Rating records used	32,51
Main tools	Python, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

The project explores a Zomato restaurant dataset to understand patterns in restaurant ratings, locations, cuisines, price ranges and selected restaurant attributes. The analysis is mainly exploratory: the purpose is to find useful patterns in the data and present them in a simple way.

2. Project Objective

The notebook was used to inspect the raw data, clean common data-quality problems, remove duplicate records, convert important fields into usable numeric form, and then study rating, location, cuisine and price patterns.

3. Dataset Description

The dataset contains restaurant-level information such as restaurant name, address, online ordering, table booking, rating, votes, location, restaurant type, liked dishes, cuisines, approximate cost for two people and listing type

Field	Observation from notebook
Rows	56,252
Columns	13
Duplicate Rows	15,703
Rows after duplicate removal	40,549
Unique restaurant names	11,914

Step	Notebook Result
Duplicate records found	15,703
Rows remaining	40,549
Rating records	32,510
Cost records	36,183
Largest missing fields	Dish_liked_50.18%

4. Data Cleaning

The notebook follows a straightforward cleaning sequence. It first checks the shape, data types, missing values and duplicate rows. Duplicate records are then removed. Column names are standardized to lowercase with underscores, and numeric fields such as votes and approximate cost are converted from text to numbers. For the rating field, numeric text is extracted and converted to numeric column. Categorical fields are stripped of extra spaces, and placeholder values such as blank strings, None, null, '-' and NEW are replaced with missing values.

5. Rating Analysis

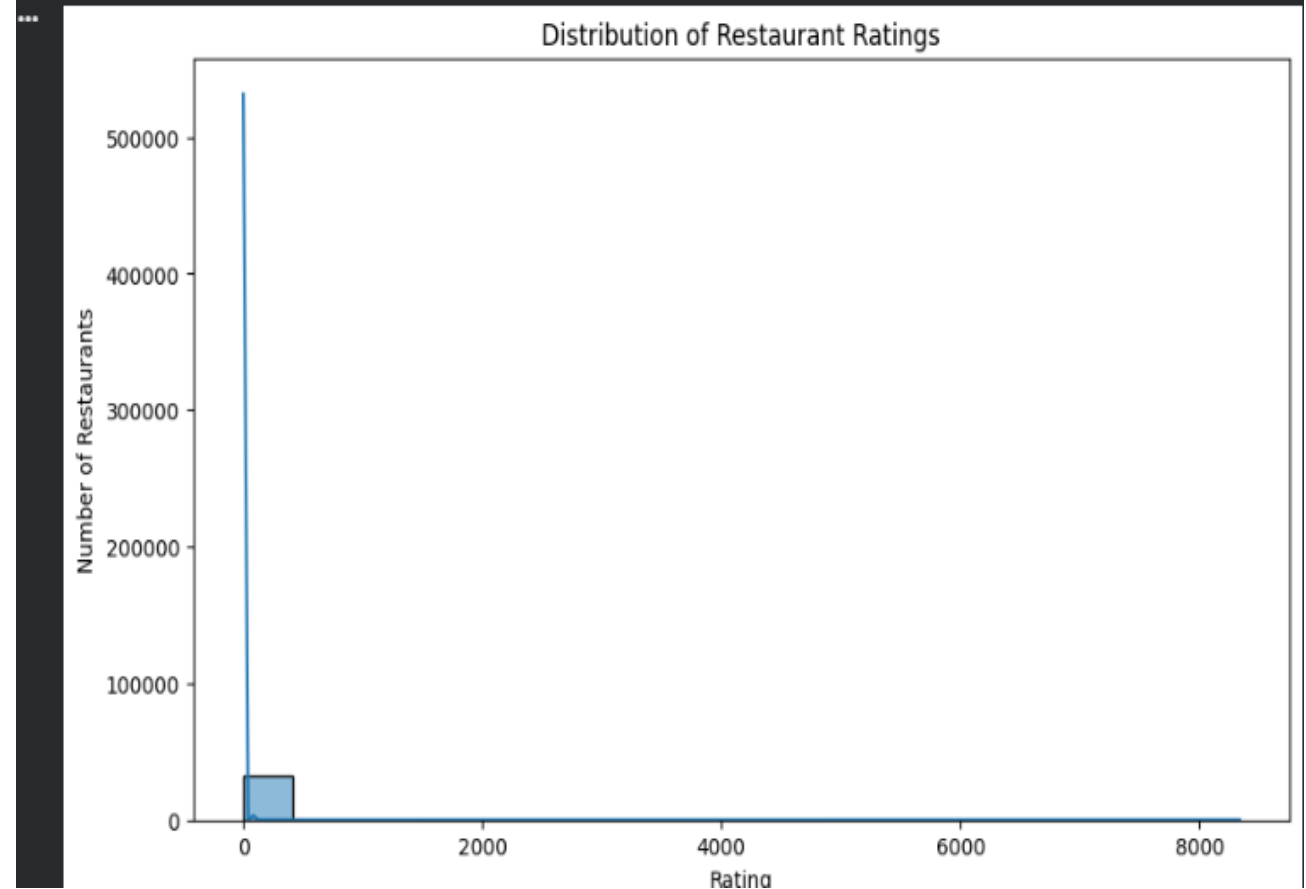
The notebook creates four rating categories: Poor, Average, Good and Excellent. The saved output contains 367 Poor records, 8,353 Average records, 12,960 Good records and 10,830 Excellent records.

```
#Rating distribution
plt.figure(figsize=(10, 6))

sns.histplot(
    rating_df["rate"],
    bins=20,
    kde=True
)

plt.title("Distribution of Restaurant Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Restaurants")

plt.show()
```



6. Location Analysis

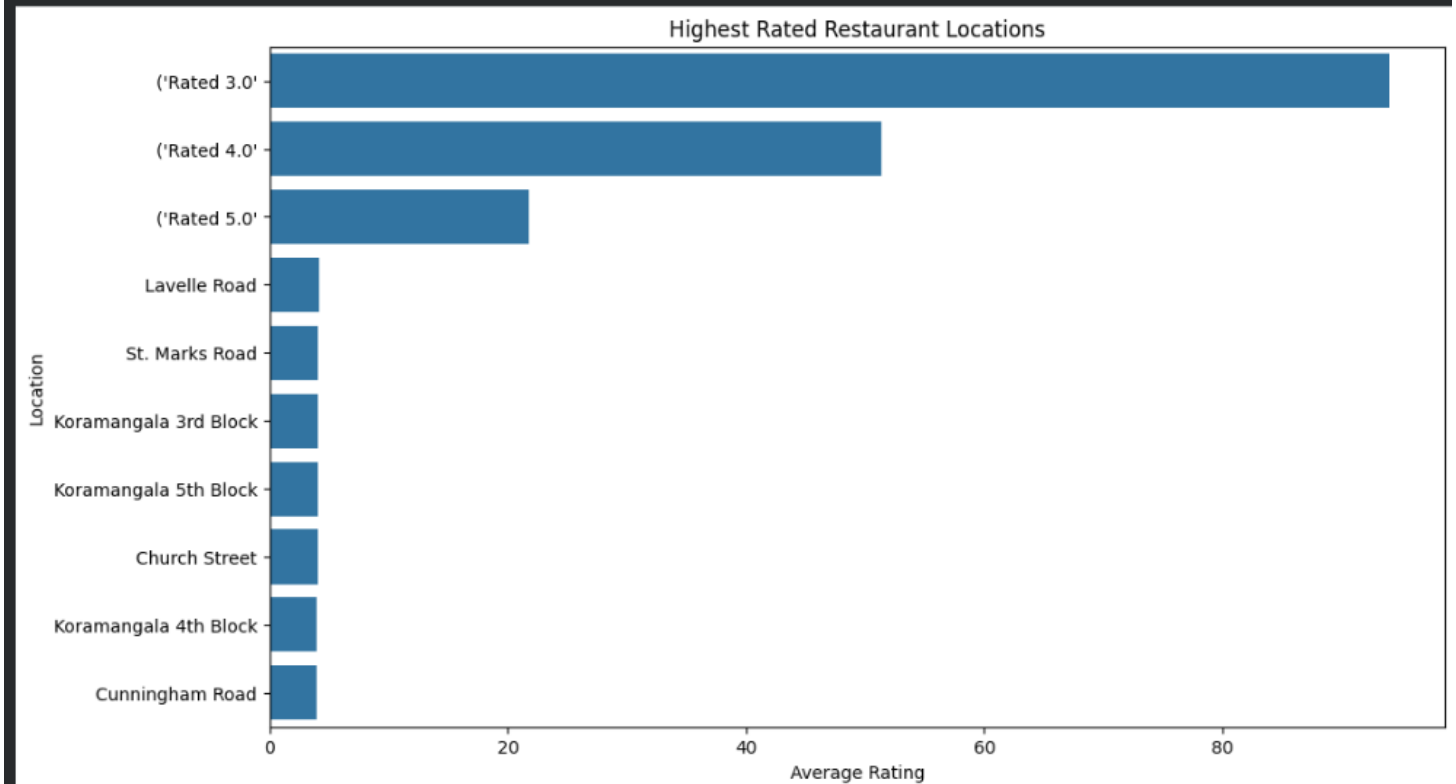
Restaurant counts are concentrated in several Bengaluru locations. BTM has the highest count in the notebook, followed by Whitefield, HSR, Indiranagar and Marathahalli.

```
#Visualization:
plt.figure(figsize=(12, 7))

sns.barplot(
    data=location_rating.head(10),
    x="average_rating",
    y="location"
)

plt.title("Highest Rated Restaurant Locations")
plt.xlabel("Average Rating")
plt.ylabel("Location")

plt.show()
```



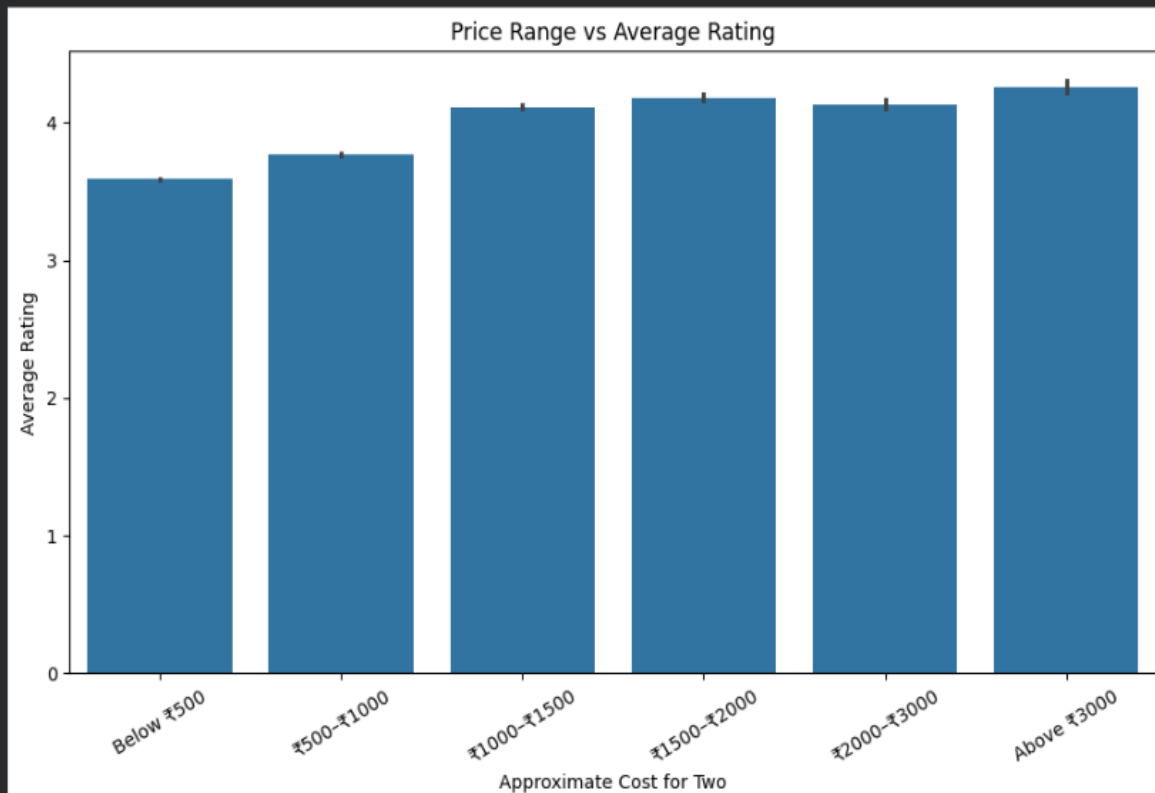
```
#Visualization:
plt.figure(figsize=(11, 6))

sns.barplot(
    data=price_df,
    x="price_range",
    y="rate",
    estimator="mean"
)

plt.title("Price Range vs Average Rating")
plt.xlabel("Approximate Cost for Two")
plt.ylabel("Average Rating")

plt.xticks(rotation=30)

plt.show()
```



7. Price vs Rating

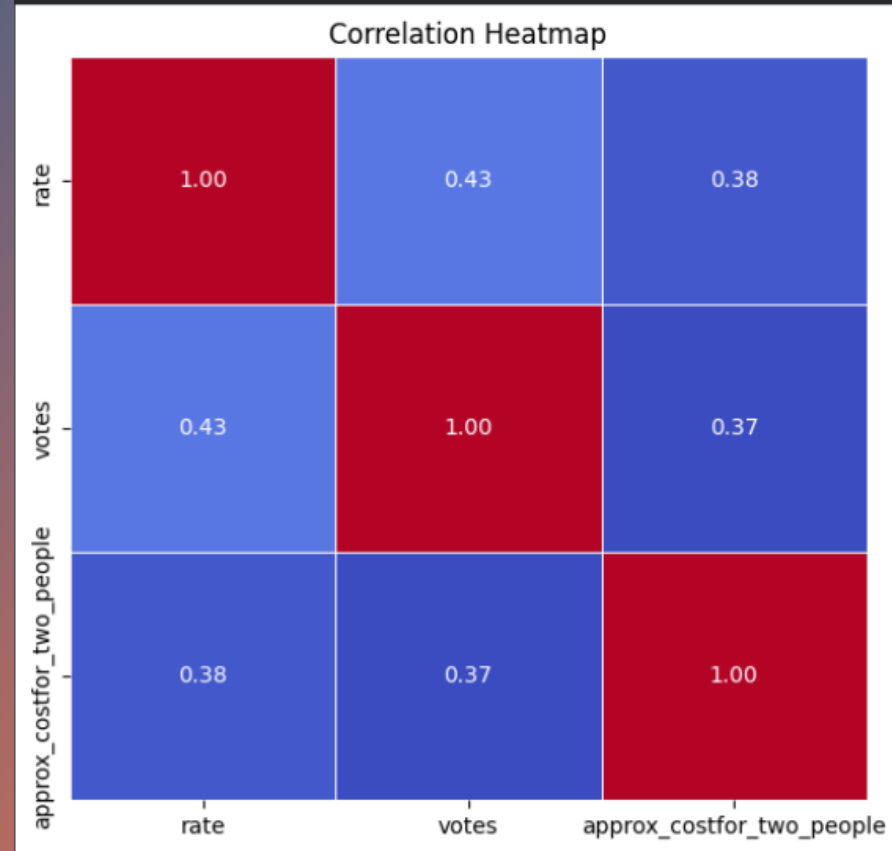
Restaurants are grouped by approximate cost for two. The average rating rises from 3.59 in the Below N500 group to 4.26 in the Above n3000 group, with some variation in the middle groups

```
#Heatmap
plt.figure(figsize=(8, 6))

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("Correlation Heatmap")

plt.show()
```



8. Correlation Analysis

The notebook calculates Pearson correlation between rating, votes and approximate cost for two people.

9. Business Recommendations

- 1. Location-based discovery:** Build local restaurant collections using restaurant density and available rating information.
- 2. Cuisine-based recommendations:** Create cuisine-focused sections for popular categories and combine them with location and price filters.
- 3. Price segmentation:** Group restaurants into budget, mid-range and premium categories to make browsing easier.
- 4. Review engagement:** Display rating and vote information together so users can see both the score and the level of user activity.
- 5. Partnership opportunities:** Use cuisine demand and location concentration to identify restaurant segments that may be relevant for promotional or partnership activities.

10. Conclusion

The project demonstrates a practical EDA workflow from loading and cleaning the dataset to extracting business-oriented observations. The analysis covers the areas requested in the notebook: rating distribution, location hotspots, cuisine popularity, price versus rating and correlation analysis. One useful lesson from this project is that successful analysis depends on data validation as much as visualization. The unusual rating values and malformed categorical outputs show that a second cleaning pass would improve the reliability of the numerical comparisons. Apart from that limitation, the notebook provides a clear starting point for understanding restaurant patterns in the dataset.